Muhammad Ammar

Machine Learning Engineer

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PROFILE

Machine Learning Engineer with hands-on experience building end-to-end machine learning solutions across classification, regression, and deep learning. Experienced in data preprocessing, feature engineering, model evaluation, and model optimization using Python and Scikit-learn. Also experienced in integrating ML models into FastAPI applications and building practical ML systems from data and modeling to inference.

EXPERIENCE

- **Machine Learning Intern** , Cellula Technology — Oct 2025 – Dec 2025 (Project-based)
  - **Hotel Booking Cancellation Prediction:** Built a classification pipeline on **500K+** hotel booking records, achieving **87% accuracy** and **0.91 F1-score** on the minority class.
  - **Uber Fare Prediction:** Built a regression pipeline on **1M+** trip records, achieving R^2 **of 0.74** and **MAE of 1.01** using gradient boosting.
  - Developed both projects end-to-end, covering **data cleaning, EDA, feature engineering, model training, and evaluation**, and integrated each model into a **FastAPI REST API** with a lightweight HTML interface for interactive local predictions.

PROJECTS

- **Real-Time Industrial Quality Control System**: **University Graduation Project**
  Stack: YOLO26n, OpenCV, ONNX, TensorRT, FastAPI, WebRTC, Firebase
  - Built a two-stage object detection pipeline using **YOLO26n**, first localizing products and then detecting defects within cropped product regions, achieving **99.4% validation mAP50** for product detection and **94.2% validation mAP50** for defect detection.
  - Exported PyTorch models through **ONNX** and implemented dynamic-shape TensorRT conversion, while using asynchronous frame processing for camera inference.
  - Integrated the detection pipeline into a **FastAPI** service with annotated video streaming, Firebase event storage, and a monitoring dashboard.
- **NYC Taxi Trip Duration Prediction**
  Stack: Python, Scikit-learn, Pandas, NumPy, FastAPI
  - Built a regression pipeline for predicting NYC taxi trip duration from **1M+ training records**, using geospatial and temporal trip features.
  - Engineered **Haversine distance** and time-based features, applied log transformations and degree-3 polynomial features, and trained a **Ridge regression** model with scaling.
  - Integrated the trained model into a **FastAPI REST API** for local trip-duration inference.
- **Road Accident Risk Prediction**
  Stack: Python, Scikit-learn, HistGradientBoosting, XGBoost
  - Built an end-to-end regression pipeline to predict road accident risk across **517K+ records**, comparing Linear Regression, Gradient Boosting, HistGradientBoosting, AdaBoost, and XGBoost.
  - Improved performance over a Linear Regression baseline from **0.804 to 0.887 test** R^2 using HistGradientBoosting with one-hot encoding and engineered numerical/categorical features.
  - Tuned model hyperparameters using **RandomizedSearchCV with 5-fold cross-validation** and implemented reusable model serialization with batch and single-record inference.

- **AI Body Measurement System**

  Stack: TensorFlow, Keras, Computer Vision, FastAPI, Python

  - Built a multi-input deep learning model combining **front and side body images** with gender, height, and weight to predict **14 body measurements**.
  - Developed a dual-branch CNN regression architecture and achieved a recorded **validation MAE of 2.86** across the predicted measurements.
  - Integrated image preprocessing, background removal, and model inference into a **FastAPI** application with a browser-based interface for measurement and clothing-size recommendations.

SKILLS

- **ML & AI:** Supervised Learning, Classification, Regression, Deep Learning, Feature Engineering, Model Evaluation, Hyperparameter Tuning
- **Frameworks & Technologies:** PyTorch, FastAPI, OpenCV, YOLO, MediaPipe
- **Libraries:** NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
- **Languages & Tools:** Python, SQL, Docker, Git, GitHub, Linux
- **Languages:** English, Arabic

COURSES & CERTIFICATIONS

- **deeplearning.ai:** Machine Learning Specialization · Deep Learning Specialization · Machine Learning in Production
- **CSkilled:** Machine Learning Diploma

EDUCATION

Mansoura University | Faculty of Computer and Information Science 2022 – 2026

Bachelor's Degree in Computer Science

LANGUAGES

- English
- Arabic